

FrameX × SBB Open Data — Test Queries

These test queries help you develop and check your rule-based system. Each one has:

- a **natural-language question**
- a **FrameX query**
- the **reference solution** from our reference system
- the **SBB dataset(s)** the answer comes from, as listed on data.sbb.ch

Disclaimer: your queries and results may look different.

The FrameX queries below were written for our reference system. How you write the same query in your own system depends on how you built it: your class and property names (e.g. `hasWifi`, `atStopPoint`, `nonStopTo`), how you name identifiers (e.g. `sp_8503000`), your semantics (open vs. closed world, how you treat cancelled trains or planned facilities), and how you define derived classes (e.g. `LongPlatform`, `Junction`, `LongDistanceStation`, `Interchange`).

Use the reference solutions as a **guide, not an exact target**. Generated IDs such as `platform_3107...` are internal to our system and will not match yours. The entities and values (stations, lengths, counts) are what should match.

Live data: "Target/Actual Comparison SBB departure/arrival times: (previous day)" only ever contains the **previous day**. It is replaced every morning. The reference solutions for those queries are based on **27 September 2026**, so your results on another day may differ.

Datasets used

All datasets are on data.sbb.ch. Search for the title to find them.

Dataset	Used in
Wifi@Station	1.1, 1.7, 2.1
Stop: platform length (body)	1.2, 2.2
Stop: waiting rooms	1.3, 2.4, 2.5
Target/Actual Comparison SBB departure/arrival times: (previous day)	1.4, 1.6, 1.7, 3.3, 3.4
Ein- und Aussteigende an Bahnhöfen (= passengers boarding and alighting at stations)	1.5, 2.6, 3.2
Service Points (Didok) based on opentransportdata.swiss	1.3, 2.1, 2.4, 2.5, 3.1, 3.5
Stop: sector boards	2.3
Line (Operation Points)	2.6, 3.1
SBB's route network	3.1

Level 1: Single dataset lookups

1.1 Does Chur have WiFi?

```
?- sp_8509000[hasWifi -> ?W].
```

Solution

```
W = true
```

Dataset: "Wifi@Station"

1.2 List every platform at Zürich HB with its length.

```
?- ?P[atStopPoint -> sp_8503000] AND ?P:Platform AND ?P[platformLength -> ?L].
```

Solution (9 platforms)

```
L = 418.0, P = platform_3107450b7a1078509da548dbb7c1f2c7
L = 424.0, P = platform_daee0a2635bc7d4e0de52344d80c3a65
L = 425.0, P = platform_c52665d4f06582a8c947dbff5c38cb1c
L = 425.0, P = platform_d91f8485fc1cc4ca6339b86e7e4ffd65
L = 426.0, P = platform_6c69110a0bbc54aeb45ee8e69fd1ac00
L = 426.0, P = platform_d530ee5465f4d066c26d56f80625ad78
L = 427.0, P = platform_928d1e02cae5d6fe595a8715843046e6
L = 428.0, P = platform_bddf15120d8baeee1844c4483b3307ae
L = 433.0, P = platform_68df83a05780d4d6ffff696d69d392f0
```

Dataset: "Stop: platform length (body)"

Note: Each result is one physical platform, not one track: island platform "10/11" serves two tracks. The underground stations Löwenstrasse (8516144) and Museumstrasse (8515163) are separate stop points, so they are not included.

1.3 Which stations in Graubünden have a waiting hall?

```
?- ?S[inCanton -> canton_gr] AND ?S[hasWaitingHall -> true] AND ?S[designation -> ?N].
```

Solution

```
N = "Chur",      S = sp_8509000
N = "Landquart", S = sp_8509002
N = "Maienfeld", S = sp_8509003
```

Datasets: "Stop: waiting rooms". The canton comes from its `kanton` field, or from "Service Points (Didok) based on opentransportdata.swiss".

1.4 Which train categories (IC, IR, S...) actually stopped in Bern?

```
?- sp_8507000[servedByCategory -> ?C].
```

Solution (operating day 27 Sept 2026)

```
C = "EC"
C = "IC"
C = "ICE"
C = "IR"
C = "TER"
```

Dataset: "Target/Actual Comparison SBB departure/arrival times: (previous day)"

Note: "Actually stopped" means excluding cancelled trains (`faellt_aus_tf`) and trains passing through (`durchfahrt_tf`). The dataset only contains SBB trains, which is why there is no S-Bahn.

1.5 Which stations had more than 50,000 daily passengers in 2024?

```
?- ?S[observedFrequency("2024") -> ?V] AND ?V > 50000 AND ?S[designation -> ?N].
```

Solution (12 stations)

```

N = "Basel SBB",      S = sp_8500010, V = 100500.0
N = "Bern",          S = sp_8507000, V = 177800.0
N = "Genève",        S = sp_8501008, V = 80600.0
N = "Lausanne",      S = sp_8501120, V = 102800.0
N = "Luzern",        S = sp_8505000, V = 103500.0
N = "Olten",         S = sp_8500218, V = 78900.0
N = "Winterthur",    S = sp_8506000, V = 101800.0
N = "Zürich Flughafen", S = sp_8503016, V = 51100.0
N = "Zürich HB",     S = sp_8503000, V = 410700.0
N = "Zürich Hardbrücke", S = sp_8503020, V = 52400.0
N = "Zürich Oerlikon", S = sp_8503006, V = 84600.0
N = "Zürich Stadelhofen", S = sp_8503003, V = 73000.0

```

Dataset: "Ein- und Aussteigende an Bahnhöfen"

Note: The solution uses **DTV** (`dtv_tjm_tgm`, the average over all days of the week). With the weekday average (DWV), Biel/Bienne would also qualify. The values count people boarding plus alighting, not unique passengers.

1.6 Where could a train take me from Bern with no stop in between?

```
?- sp_8507000[nonStopTo -> ?B] AND ?B[designation -> ?N].
```

Solution (operating day 27 Sept 2026)

```

B = sp_8500218, N = "Olten"
B = sp_8502001, N = "Zofingen"
B = sp_8503000, N = "Zürich HB"
B = sp_8504100, N = "Fribourg/Freiburg"
B = sp_8504103, N = "Flamatt"
B = sp_8504414, N = "Lyss"
B = sp_8507006, N = "Münsingen"
B = sp_8507100, N = "Thun"
B = sp_8508005, N = "Burgdorf"

```

Dataset: "Target/Actual Comparison SBB departure/arrival times: (previous day)"

1.7 Which long-distance stations have no WiFi? (*open-world test*)

```
?- ?S:LongDistanceStation AND NOT ?S[hasWifi -> true].
```

Solution

```
unknown
```

Datasets: "Wifi@Station" + "Target/Actual Comparison SBB departure/arrival times: (previous day)"
(to define `LongDistanceStation`)

Note: The expected answer is `unknown`. The world is open and no dataset is declared complete for `hasWifi`, so a missing WiFi record does not prove that a station has no WiFi. This query shows how the open-world assumption works.

Level 2: Joins across datasets

2.1 Which stations in Ticino have WiFi?

```
?- ?S[inCanton -> canton_ti] AND ?S[hasWifi -> true] AND ?S[designation -> ?N].
```

Solution

```
N = "Bellinzona", S = sp_8505213  
N = "Locarno",    S = sp_8505400  
N = "Lugano",     S = sp_8505300
```

Datasets: "Wifi@Station" + "Service Points (Didok) based on opentransportdata.swiss" (for the canton)

Note: "Wifi@Station" has no canton field. Join on the stop-point number (`bpuic` = `number`).

2.2 Which platforms at Bern are longer than 320 m? Give the platform number and length.

```
?- ?P:LongPlatform AND ?P[atStopPoint -> sp_8507000] AND ?P[platformNumber -> ?No] AND  
?P[platformLength -> ?L].
```

Solution (7 platforms)

```
L = 362.0, No = "12/13", P = platform_b59ffde62ae2b782e4228b7995c02c9a
L = 368.0, No = "49/50", P = platform_14af45e646ab9eccbd45b83b27e9e918
L = 369.0, No = "9/10", P = platform_9b962c759b1d713b387339316f46fc01
L = 506.0, No = "5/6", P = platform_aa0b04fdfb0ac2ef34939d52dfa6ef1f
L = 510.0, No = "3/4", P = platform_dd92174a5181ae5452c1f64da6eb73fd
L = 521.0, No = "7/8", P = platform_695594f9869c0fdc5122290b8c9855ed
L = 539.0, No = "1/2", P = platform_934726eff0382c9ab3c5bf70cee6e73c
```

Dataset: "Stop: platform length (body)"

Note: `LongPlatform` is a derived class ($\text{length} > 320 \text{ m}$). All platforms listed for Bern exceed that threshold.

2.3 Which sector letters are signed on track 3 at Zürich HB?

```
?- ?B:SectorBoard AND ?B[atStopPoint -> sp_8503000] AND ?B[trackNumber -> "3"] AND ?B[sectorFront -> ?F].
```

Solution (7 boards, letters A–D)

```
B = sectorboard_525d71545e26c6099d1a220fa267360b, F = "D"
B = sectorboard_646f9a6a8d5844a05df1003dd1d38052, F = "C"
B = sectorboard_880862013b865aa5590c5971378e2a8d, F = "B"
B = sectorboard_9ad31eedce6a29c6897a6ef0ae786d5e, F = "C"
B = sectorboard_b062e6a23af4293b7a022e2efaba138d, F = "A"
B = sectorboard_d13c933a04fdafd72d1c167e55727ae9, F = "B"
B = sectorboard_d3f85a02213933b0bee6f521f11a3c29, F = "A"
```

Dataset: "Stop: sector boards"

2.4 Which stations in canton Bern have a waiting hall that is planned but not built yet?

```
?- ?F:WaitingHall AND ?F[status -> "PROJEKTIERT NEU"] AND ?F[atStopPoint -> ?S] AND ?S[inCanton -> canton_be] AND ?S[designation -> ?N].
```

Solution (8 stations)

```

F = waitinghall_0702bde03f234f24b9fe4791190ec8c2, N = "Brügg BE",      S = sp_8504416
F = waitinghall_106c0b5e73262bbc2ac1b6550172cc35, N = "Kallnach",      S = sp_85044
02
F = waitinghall_124c0226864b9873a480969a722f3d49, N = "Neuenegg",      S = sp_85041
92
F = waitinghall_6685757fb1a47f1fc2fd62c76d9c6e21, N = "Uttigen",      S = sp_85070
09
F = waitinghall_c118077c4fa1267fd15c40e9e42ab45a, N = "Twann",      S = sp_85042
28
F = waitinghall_cc5e644d7879743f82f310a0dc8700fc, N = "Herzogenbuchsee", S = sp_85080
08
F = waitinghall_e0f9c521c18b9c6481c8f259ee2a65e6, N = "Hindelbank",      S = sp_85080
03
F = waitinghall_f95cebbb03869ff2d424af958450fafb, N = "St-Imier",      S = sp_85043
10

```

Datasets: "Stop: waiting rooms". The canton comes from its `kanton` field, or from "Service Points (Didok) based on opentransportdata.swiss".

2.5 Which waiting halls in canton Zürich are planned for demolition?

```

?- ?F:WaitingHall AND ?F[status -> "PROJEKTIERT ABBRUCH"] AND ?F[atStopPoint -> ?S] AND
  ?S[inCanton -> canton_zh] AND ?S[designation -> ?N].

```

Solution (8 halls at 7 stations)

```

F = waitinghall_327e2ca4df41dfddbc616ba14c261da4, N = "Wallisellen",      S = sp_8503129
F = waitinghall_7e77303fdae9125b1fe7ceel6b8ff95f, N = "Stäfa",          S = sp_850310
7
F = waitinghall_9448cb1fe0e4a3f027dc2b474869dab3, N = "Zürich Seebach", S = sp_850300
7
F = waitinghall_adcd16a7c79a4da1d0a0929282b00143, N = "Stäfa",          S = sp_850310
7
F = waitinghall_af33de5ed54d665dlb05e0c35beb94e5, N = "Dietlikon",      S = sp_850330
6
F = waitinghall_bbdc6f9cedab81c878c3c47aa9172752, N = "Bassersdorf",    S = sp_850330
7
F = waitinghall_c65fff2227625e3f925b86660c88dd13, N = "Stettbach",      S = sp_850314
7
F = waitinghall_cb616c4a15f50eb9a26c8b3407ec8ee3, N = "Aathal",        S = sp_850312
4

```

Datasets: "Stop: waiting rooms". The canton comes from its `kanton` field, or from "Service Points (Didok) based on opentransportdata.swiss".

2.6 How many people used each station on line 900 in 2024?

```

?- ?S[servedByLine -> line_900] AND ?S[observedFrequency("2024") -> ?V] AND ?S[designat
ion -> ?N].

```

Solution

```

N = "Bad Ragaz", S = sp_8509004, V = 3100.0
N = "Chur",      S = sp_8509000, V = 28500.0
N = "Landquart", S = sp_8509002, V = 18600.0
N = "Maienfeld", S = sp_8509003, V = 980.0

```

Datasets: "Line (Operation Points)" + "Ein- und Aussteigende an Bahnhöfen"

Note: Zizers SBB is on line 900 but has no 2024 passenger count, so it drops out of the join.

Level 3: Derived classes and comparisons

3.1 Which stations are junctions in Graubünden, and which lines meet there?

```
?- ?S:Junction AND ?S[inCanton -> canton_gr] AND ?S[servedByLine -> ?L] AND ?L[label -> ?LL] AND ?S[designation -> ?N].
```

Solution: Chur (lines 900, 920) and Landquart (lines 900, 910, 920)

```
L = line_900, LL = "900", N = "Chur",      S = sp_8509000
L = line_900, LL = "900", N = "Landquart", S = sp_8509002
L = line_910, LL = "910", N = "Landquart", S = sp_8509002
L = line_920, LL = "920", N = "Chur",      S = sp_8509000
L = line_920, LL = "920", N = "Landquart", S = sp_8509002
```

Datasets: "Line (Operation Points)" + "Service Points (Didok) based on opentransportdata.swiss" (for the canton) + "SBB's route network" (for line labels)

> **Note:** **Junction** = a stop point served by at least 2 lines.

3.2 Which stations were busy in 2025 (over 20,000 passengers a day) but had 20,000 or fewer in 2018?

```
?- ?S[busyIn("2025") -> true] AND ?S[observedFrequency("2018") -> ?V] AND ?V <= 20000 AND ?S[designation -> ?N].
```

Solution (V = 2018 value)

```
N = "Arth-Goldau", S = sp_8505004, V = 14200.0
N = "Bellinzona", S = sp_8505213, V = 16000.0
N = "Brig",        S = sp_8501609, V = 16700.0
N = "Lugano",      S = sp_8505300, V = 16300.0
N = "Renens VD",   S = sp_8501118, V = 18300.0
N = "Schaffhausen", S = sp_8503424, V = 18900.0
```

Dataset: "Ein- und Aussteigende an Bahnhöfen"

Note: `busyIn(Year)` is a parameterised derived property (DTV > 20,000 in that year).

3.3 Which long-distance stations can you reach from Zürich HB without stopping?

```
?- sp_8503000[nonStopTo -> ?B] AND ?B:LongDistanceStation AND ?B[designation -> ?N].
```

Solution (18 stations, operating day 27 Sept 2026)

```
B = sp_8500010, N = "Basel SBB"
B = sp_8500218, N = "Olten"
B = sp_8500309, N = "Brugg AG"
B = sp_8502113, N = "Aarau"
B = sp_8502119, N = "Lenzburg"
B = sp_8502204, N = "Zug"
B = sp_8502206, N = "Baar"
B = sp_8503001, N = "Zürich Altstetten"
B = sp_8503006, N = "Zürich Oerlikon"
B = sp_8503016, N = "Zürich Flughafen"
B = sp_8503202, N = "Thalwil"
B = sp_8503206, N = "Wädenswil"
B = sp_8503424, N = "Schaffhausen"
B = sp_8503504, N = "Baden"
B = sp_8506000, N = "Winterthur"
B = sp_8506302, N = "St. Gallen"
B = sp_8507000, N = "Bern"
B = sp_8509411, N = "Sargans"
```

Dataset: "Target/Actual Comparison SBB departure/arrival times: (previous day)"

Note: `LongDistanceStation` = a station served by at least one long-distance category (IC, IR, EC, ICE, TGV, NJ, RJX...). Winterthur and St. Gallen appear only because of a single night train (NJ).

3.4 Which stations had TGV service?

```
?- ?S[servedByCategory -> "TGV"] AND ?S[designation -> ?N].
```

Solution (operating day 27 Sept 2026)

```
N = "Basel SBB", S = sp_8500010
N = "Genève", S = sp_8501008
N = "Lausanne", S = sp_8501120
N = "Vallorbe", S = sp_8501103
N = "Zürich HB", S = sp_8503000
```

Dataset: "Target/Actual Comparison SBB departure/arrival times: (previous day)"

Note: The dataset also contains TGV stops in France (Paris Gare de Lyon, Dijon, Mulhouse...). The reference solution lists only the Swiss ones.

3.5 Which stations are interchanges to the tram?

```
?- ?S:Interchange AND ?S[servesMode -> mode_tram] AND ?S[designation -> ?N].
```

Solution (6 stations)

```
N = "Worb Dorf", S = sp_8507063
N = "Zürich Stadelhofen, Bahnhof", S = sp_8503059
N = "Zürich, Balgrist", S = sp_8530811
N = "Zürich, Hegibachplatz", S = sp_8530812
N = "Zürich, Rehalp", S = sp_8591315
N = "Zürich, Wetlistrasse", S = sp_8591429
```

Dataset: "Service Points (Didok) based on opentransportdata.swiss"

Note: `Interchange` = a service point whose `meansoftransport` contains both `TRAIN` and `TRAM`. Big stations such as Zürich HB are not included because their tram stops are separate service points.